Mahmoud Alyosify

Machine Learning Engineer | End-to-End ML Pipelines, Training & Deployment
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PROFESSIONAL SUMMARY

Machine Learning Engineer who takes models from **raw data to a deployed, monitored service**: feature engineering, training, systematic hyperparameter search, honest evaluation, and packaging for production via **ONNX, Docker, FastAPI and AWS**. Currently an **AI Engineer Intern at Injaz Tech**, turning an LLM/RAG prototype into a tested **FastAPI** service with a CI evaluation gate. Completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship**. Delivered supervised, self-supervised and reinforcement-learning systems — including a **42-experiment** vision sweep reaching **84.30% top-1 accuracy** with ONNX/FAISS retrieval, a satellite collision-risk model operating on a **48-hour** decision window, and a fine-tuned LLM served on AWS. Strong software-engineering base from national-scale backend systems, and **6 years** teaching ML to **10,000+ learners**.

EDUCATION

MSc in Artificial Intelligence — *Sep 2025 – Nov 2026*
Queen's University, Kingston, Ontario, Canada — Expected Nov 2026 · GPA: 4.30 / 4.30

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- Coursework: **Deep Learning, Large Language Models, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines**.
- **Master's Thesis:** *MinimalLM: Verbosity-Aware Decoding for Token-Efficient Language Model Inference* — Supervisor: Dr. Rahatara Ferdousi.

B.Sc. Computer & Information Science (Bioinformatics) — *Sep 2019 – Jul 2023*
Assiut University, Egypt — GPA: 3.53 / 4.00

- Graduation Project: **Vitalism Solution** — contactless vital-sign estimation via rPPG. **Grade A+**; **Semi-Finalist, Microsoft Imagine Cup 2023**.

TECHNICAL SKILLS

ML & Deep Learning	PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, CNNs, Self-Supervised & Contrastive Learning, Reinforcement Learning (PPO, DQN), Transformers, Fine-Tuning (LoRA, QLoRA), RAG (LangChain, LangGraph)
Modelling Workflow	Feature Engineering, Class-Imbalance Handling, Cross-Validation, Hyperparameter Optimisation (Optuna), Model Selection, Explainability (SHAP), Ablation Studies, Error Analysis
Data Engineering	Pandas, NumPy, PySpark, ETL Pipelines, SQL, Data Cleaning & Validation, MySQL, MongoDB, SQL Server
Deployment & MLOps	ONNX, Docker, FastAPI, Streamlit, AWS, CI/CD (GitHub Actions), pytest, FAISS, ChromaDB, Git/GitHub, Linux, Reproducible Experiment Design, Latency & Resource Profiling
Programming	Python, SQL, C++, C, C#, Go, JavaScript, MATLAB, Bash

PROFESSIONAL EXPERIENCE

AI Engineer Intern — *Oct 2026 – Present*
Injaz Tech — Injaz Group (formerly Fingerprint Consultancy) — Maadi, Cairo, Egypt

- Built an **agentic RAG** assistant (**LangChain, LangGraph**, FAISS, cross-encoder re-ranking) served via **FastAPI** with **Docker**, **136 pytest** tests and a CI evaluation gate.
- Raised top-1 retrieval accuracy from **75% to 94%** and cut false refusals from **5/20 to 1/20** through measured design changes.

Part-Time Instructor — Machine Learning & AI for Business — *Jul 2025 – Present*
National Telecommunication Institute (NTI) — Egypt

- Teach **Machine Learning** (90-hour bootcamps) and **AI for Business** modules in NTI's Summer Internships Program, on separate engagements.

AI in Cybersecurity Intern — *Jun 2025 – Jul 2025*
e& Egypt — via NTI (On-the-Job Training) — Cairo, Egypt

- Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) — *Feb 2024 – Mar 2025*
National Company for Roads Building & Development — Cairo, Egypt

- Supported backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles** across **70%+ of national roads**.
- Co-developed an internal data-management and reporting desktop application in **C#, .NET, Entity Framework, SQL Server** and **Crystal Reports**.

Online Instructor — AI, Algorithms & Data Science *2020 – Present*
Udemy & YouTube Remote

- Built and taught Arabic-language technical curricula reaching **10,000+ learners** and **700K+ YouTube views**.

SELECTED MACHINE LEARNING PROJECTS

SimCLR-Vision-SSL — Self-Supervised Learning with Deployment-Ready Retrieval *2026*
PyTorch · SimCLR · SupCon · ONNX · FAISS

- Built a full self-supervised training pipeline and ran a **42-experiment augmentation sweep** to isolate which transformations actually drive representation quality.
- Extended the model with a semi-supervised **SupCon** objective and reached **84.30% top-1 accuracy**, evaluated on both classification and image-retrieval tasks.
- Exported the trained encoder to **ONNX** and served embeddings through a **FAISS** index, making the pipeline deployment-ready rather than notebook-only.

SCRAP — Satellite Collision Risk Assessment & Prediction *2026*
Python · Scikit-learn · Supervised Learning · Feature Engineering

- Built a supervised ML pipeline that predicts satellite collision risk using only conjunction data available **48+ hours** before closest approach — deliberately constrained to match the real operational decision window rather than maximising offline accuracy.
- Handled a heavily imbalanced target and engineered features from raw conjunction data, evaluating with metrics appropriate to rare, high-cost events.

RL-Cloud-Autoscaler — Deep RL for Cloud Resource Provisioning *2026*
Python · PPO · DQN · Gymnasium

- Built a **custom Gymnasium environment** simulating cloud workload dynamics and trained **PPO** and **DQN** agents to optimise the **cost–latency trade-off**, benchmarked against static and threshold-based autoscaling baselines.

Horus-OSINT — Fine-Tuned LLM Served on AWS *2026*
Meta-Llama-3-8B · Fine-Tuning · AWS · Python

- Fine-tuned **Meta-Llama-3-8B** on **159,826** GTD/GDELT records — dataset construction through training and evaluation — and deployed it on **AWS** as an interactive assistant.

Additional Projects

- **AudioShield** — CNN trained on mel-spectrograms for deepfake audio detection, served with real-time inference through a Streamlit application.
- **MinimalLM** — MSc thesis: Fewer generated tokens are a real saving only when the answer is still complete. MinimalLM gives both, without training. A training-free decoding method makes three open LLMs generate **47.6%** fewer tokens and answer **19–68%** faster, with no answer cut off.
- **Human vs. Agentic Pull Requests** — reproducible GitHub-mining and analysis pipeline over large-scale open-source repositories.
- **Also:** Vitalism Solution (rPPG) · Diabetes Onset Prediction · ATM Transaction Analysis · Automated Multimodal Agent.

CERTIFICATIONS & AWARDS

AWS Academy — ML Foundations, ML for NLP, Data Engineering
HCIP – AI (Huawei)
ML Internship — ITIDA / Egypt Makes Electronics (200 hrs)
1st Place — Smart Cities Hackathon 2022
Machine Learning Specialization — DeepLearning.AI
Microsoft Certified: Data Analyst Associate
AI in Cybersecurity — NTI (420 hrs)
Semi-Finalist — Microsoft Imagine Cup 2023

LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency.